

OM KEKAN

AI/ML Engineer · Generative AI & LLM Systems

Chhatrapati Sambhajnagar, Maharashtra, India | +91 8788882314 | omkekan27@gmail.com | github.com/Omkekan | linkedin.com/in/omkekan

PROFILE

AI/ML Engineer with hands-on, end-to-end machine learning experience from data preprocessing with Pandas and NumPy to deploying agentic GenAI pipelines and automated analysis tools. Proficient in Python, TensorFlow, and PyTorch, with a growing specialization in Large Language Models, Retrieval-Augmented Generation (RAG), and applied Generative AI. Builds evaluation-driven systems that quantify and improve model reliability rather than relying on anecdotal testing.

EXPERIENCE

Operational Assistant Intern — Inspira Technologies

4-Month Internship

- Developed and maintained backend services in Python and Flask, building and testing API endpoints that supported core application features and internal workflows.
- Designed and implemented a chatbot module to automate responses to common user queries, integrating conversational logic with backend services to streamline routine interactions.
- Collaborated with the team on debugging, testing, and code reviews, and contributed to internal documentation to support maintainability of the systems built during the internship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, LangChain, LangGraph, RAGAS, Computer Vision, NLP, Generative AI, Large Language Models (LLMs)

Backend & Infrastructure: Flask, Next.js, Celery, Redis, Qdrant, Ollama

Databases & Tools: MySQL, PostgreSQL, Git, GitHub, Linux

Core Competencies: End-to-End ML Deployment, Vulnerability Assessment, Information Security

PROJECTS

Autonomous RAG Agent with Hallucination Guardrails | Python, LangChain, LangGraph, Qdrant, Ollama

<https://github.com/Omkekan/Auto-RAG-Agent-with-Hallucination-Guardrails>

- Engineered a self-correcting RAG pipeline in LangGraph, adding retrieval-relevance grading, automatic query rewriting, and post-generation groundedness checks to reduce hallucinated responses versus a naive RAG baseline
- Designed a closed-loop agentic architecture that detects weak retrievals and unsupported answers at runtime and automatically re-queries or regenerates before returning a response, rather than returning a first-pass answer
- Built an evaluation harness using RAGAS (faithfulness, answer relevancy) to benchmark the guarded pipeline against a baseline RAG system, replacing anecdotal testing with quantified reliability metrics

Web Vulnerability Scanner | Python, Flask, Celery, Redis, Next.js, PostgreSQL

<https://github.com/Omkekan/Riskreveal-Vulnerability-scanner>

- Built a risk-prioritization engine correlating data across 3+ security feeds (200,000+ CVEs, 1,000+ CWEs), cutting vulnerability noise by an estimated 90% versus raw scanner output
- Architected a high-performance backend with Celery and Redis supporting up to 50 concurrent scans, cutting manual scan-configuration time by over 70% through automation
- Developed a full-stack security dashboard in Next.js with 15+ reusable components, improving development efficiency by 25% and enabling detailed PDF report generation in under 5 seconds

Additional Projects

- Text Classification for News Articles — Python, Scikit-learn, Pandas — trained and evaluated an NLP classification model to categorize news articles by topic. github.com/Omkekan/Text-classification-for-News-Articals
- Network Scanner — Python, Scapy — built a multithreaded TCP port scanner using SYN scanning to identify open ports and running services on a target host. github.com/Omkekan/network-scanner

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning

Oct 2021 – Aug 2026

IICT MGPU, Chhatrapati Sambhajnagar, India

CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI (In Progress)
- Introduction to Large Language Models — Google
- Generative AI Fundamentals — Google
- Google Cybersecurity Professional Certificate